

ActInf Livestream #025.0 ~ "The Computational Boundary of a Self"

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Duration: 1:40:19 | **Uploaded:** Unknown **Transcript method:** auto_caption

foreign go for it hello everyone welcome to acton flab live stream number 25.0 today is july 1st 2021 and we're going to be talking about this paper the computational boundary of a self developmental bioelectricity drives multicellularity and skill free cognition i'm blue and i'm here with daniel awesome so welcome to the active inference lab everyone we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this page this is a recorded and archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here we will be following good video etiquette for live streams here at the short link you'll find all of the live streams and different series that we do in the communications unit of the active inference lab and today we're doing the contextualizing paper the dot zero video for two upcoming discussions in the first half of july 2021 on the 6th and the 13th when we have discussions 25.1 and 25.2 on this paper and hopefully the author joining today in active lab live stream number 25.0 we're going to be trying to set some context and give an introduction to the following paper the computational boundary of a self developmental bioelectricity drives multicellularity and scale free cognition by mike levin and this video is just an introduction to some of the ideas it's not a review or a final word it's kind of like a three-way intersection where we have people who might be within the active inference community looking to be exposed to some different areas like developmental biology or cognitive science and then another group of people who are coming from the developmental biology or cognitive science background and curious about active inference and then also we hope that this is exciting and interesting even for people who are not familiar with active inference or developmental we will hopefully try to connect it with some broader questions so we're going to walk through some of the aims and claims of the paper the abstract and the roadmap and bring a few big questions and some key points of the paper so that whether you read the paper or not you'll hopefully be in a good spot to ask questions and learn

more and of course in the 1 and 2 videos in the coming weeks we'll be discussing the same paper so save and submit your questions and let us know if you'd like to participate or contribute in any way so here we are on the paper itself that's a screenshot of the cover and the aims and claims and i'll read them so that then daniel you can give a first thought on what you thought was cool or what you thought was important about the paper so the aims and claims of the paper first they define individuals themselves in a way that facilitates taxonomy comparison and communication with evolved created biological artificial and exobiological agents he says i propose a fundamental definition of an individual based on the ability to pursue goals at an appropriate level of scale and organization and suggest a formalism for defining and comparing the cognitive capacities of highly diverse types of agents second he proposes a plausible naturalistic framework for the evolutionary scale-up of cognition from the earliest origins of life hypothesizing about the forces that drove it and the major transitions along the continuum and the goal of this research program is to show how complex agency and goal directiveness evolves naturally from ancient mechanisms so daniel what do you think about that nice work well i thought the paper was pretty exciting because it touches on just so many big topics like goal setting and agency cognition in natural and artificial systems multi-scale systems and then it links it in a way that people might not expect which is through developmental biology and bioelectricity which are kind of like processes that these life forms or other forms engage in so to be able to connect things that are awesome about systems that matter like again their goal seeking nature their ability to do cognition and then connect that to a process theory it's kind of what we're all about in active inference connecting the cool features of the world to process theories that are tractable cool so moving on to the abstract i'll read the first part and then you can read the next slide so all epistemic agents physically consist of parts that must somehow comprise an integrated cognitive self biological individuals consist of subunits organs cells and molecular networks that are themselves complex and competent in their own native contexts how to how do coherent biological individuals result from the activity of smaller sub-agents to understand the evolution and function of metazoan creatures bodies and minds it is essential to conceptually explore the origin of and the scaling of basal cognition of individual cells into a coherent larger organism so i find this line of research truly fascinating and really a burning question for me is how information is consolidated or prioritized across skills whether it's core screening or salience or something else so do you want to read the next part yep just one thought on that and it's going to be a theme that's returned to it's that cognition or agency doesn't just blink into

existence at a larger level of analysis it's actually composed of smaller cognitive units and so it's kind of like cognition all the way down that helps us integrate across scales but also recognize what's special about larger levels of organization like culture so part two of the abstract in this article i synthesize ideas in cognitive science evolutionary biology and developmental physiology towards a hypothesis about the origin of individuality scale-free cognition i propose a fundamental definition of an individual based upon the ability to pursue goals at an appropriate level of scale and organization and suggest a formalism for defining and comparing the cognitive capacities of highly diverse types of agents any self is demarcated by a computational surface the spatial temporal boundary of events that it can measure model and try to affect the surface sets a functional boundary a cognitive light cone which defines the scale and limits of its cognition i hypothesize that higher level goal-directed activity and agency resulting in larger cognitive boundaries evolves from the primal homeostatic drive of living things to reduce stress the difference between current conditions and life optimal conditions so i like this goal directed like aspect of this and it really kind of implies to me that the it's like biological cognition has to be biological but i wonder about like you know what is the goal of of molecules in their like organization right is that like a goal directed activity like the protons and neutrons and electrons want to like reduce their uncertainty about where they are in space or how does it work at like a non-biological level makes me wonder that or strain strain and stress those are things we psychologically experience but then you can have you know like a ring with four carbons it's under strain and the carbons they seek to do what well get into a less stressed position like a unstrained arrangement for sure the next part the mechanisms of the ability of all cells to form electrical networks that process information suggests a plausible set of gradual evolutionary steps that naturally lead from physiological homeostasis in single cells to memory prediction and ultimately complex cognitive agents via scale up of the basic drive of infotaxis recent data on the molecular mechanisms of pre-neural bioelectricity suggest a model of how increasingly sophisticated cognitive functions emerge smoothly from cell to cell communication used to guide embryogenesis and regeneration this set of hypotheses provides a novel perspective on numerous phenomena such as cancer and makes several unique testable predictions for interdisciplinary research that have implications not only for evolutionary developmental biology but also for biomedicine and perhaps artificial intelligence and exobiology do you have any thoughts on that daniel i guess it's just cool to see development and cognition come on to equal grounding because there was ecology evolution and development eco evo devo

and then it's sort of that's natural things happening and then often cognition and especially computational perspectives on cognition are seen as unnatural or designed and so again all of these pieces are going to be woven together with a few pieces like bioelectricity and light cones that maybe people aren't uh expecting cool so the road map of the paper we're gonna go through the introduction then the paper is broken up into lots of different parts what is the self how do you define an individual body patterning cognition multicellularity versus cancer defining individuation from a cognitive the agent's evolutionary backstory conclusion future outlook discussion and then uh some predictions and research program and then in the end uh what does it feel like to be a pancreas which i i renamed that part but maybe you changed it back so we'll see do you want to read over the keywords so here's the provided keywords were development bioelectricity and the related gap junctions primitive cognition and our favorite active inference so development i like this quote from smith seth mary and maynard smith which i can never say or says last name i feel bad sorry yours developmental biology can be seen as the study of how information in the genome is translated into the adult structure and evolutionary biology of how the information came to be there in the first place so that's a kind of cool definition of development do you have any thoughts on that it reminds me of earlier discussions we had about tin bergens four y's which are kind of related to aristotle's four whys it's like why is there a foot well the easy sort of material answer is because there's matter in the shape of a foot but maybe you want to know something that's historical and over really long periods of time it's a story about evolution how the foot came to be selected on through time but development is just right there in the middle where it's the story for that individual how that foot came to be um thanks said blue could you turn down my mic it um apparently it's a little loud or distorted thank you for the feedback yeah if anybody has comments on like the audio um blue's doing an awesome job of first broadcaster role so thanks ed cool all right hopefully that's a little bit better all right uh bioelectricity so this is a quote from the paper developmental bioelectricity is the ubiquitous exchange of slowly changing ion-based voltage signals within and among cells all cells are electrically active and modern neurons evolved from pre-neural precursors that were already reaping the benefits of ionic signaling for computation computation ionic signaling for so there's some ion channels i like that cool like digital um image of ion channels and cells and of course this painting by alex gray like isn't officially bioelectricity but whenever i think of the word bioelectricity i always think of this picture do you have any thoughts and to connect the developmental aspect to what is mentioned here about the cells being electrically active the neural tissue arises during

development from neuro ectoderm so before there was sort of an inner layer of these electrically specialized ones that we call neurons there was an outer layer and we see that recapitulated in embryogenesis when the neural tube is closed so that's combining sort of ontogeny recapitulates phylogeny which is just to say that the development of the organism carries some resonance with the way that the organism evolved through time and then just to remind that every cell has those pumps that maintain charge maybe not all of them are involved in rapid fire signaling but to maintain ionic gradients and therefore bioelectric gradients is essentially what cells do with remembrance cool and awesome segway into talking about gap junctions so this is like something that's near and dear to my heart with my like neuroscience background so gap junctions like were originally attributed to like neurons right like that's it like gap junctions are there at the synapse and you know they facilitate the electrical signaling between neural cells but but really they're um channels that permit cell to cell transfer of ions directly and they were originally described in nerve and muscle but they're really found in virtually all cells that are present in solid tissues um and then i just did a quick like brief look up of a recent review of gap junctions and it's cool like they're involved in all kinds of um things right like so self proliferation immune response migration optosis carcinogenesis hyperproliferative skin disorders lymphatic vessel diseases inflammatory lung diseases that's like asthma liver injury and neoplastic disorders that's also cancer so i thought that that was kind of a neat um thing any thoughts on this one daniel just that active inference helps us think about how agents communicate and interact so if there's two cells and they have totally disjoint membranes then they can both modify a niche in common so they could pump ions in and out of the fluid like the extracellular matrix that's surrounding the cells um so there's a common niche that they're sharing that's like the least connected they could be and then there are also ways in which they can have a long-term connection with a higher throughput so we're seeing a lot of these ideas of communication where you can talk about these two agents now directly exchanging with each other as their electrochemical niche rather than just jointly modifying the shared fluid that they're in cool okay next all right primitive cognition see if i can play this video did you want me to play it sure if it works on your side let's see it perfect role of acton in cell motility stunning 3d animation maybe we don't need the sound but um next well what um is written in the paper is the emerging field known as basal tracks the evolutionary history of learning and decision making processes beginning from the dynamic problem-solving capacities of cellular and subcellular forms many examples of memory anticipation context-dependent decision-making and learning are exhibited by

organisms from yeast and bacteria to plants and somatic cells cool so down at the bottom what does cognition or intelligent behavior look like across scales where and how does cognition arise and how does it develop or evolve convergent and divergent features and where does active inference come into play i think that first question is fun because it's a little easy to off the cuff just say well i i don't recognize the intelligence of a cell because it's not um you know doing an undergrad course or it's not doing skillful performance like we talked about in a previous discussion so will we have a human-centric conception of intelligence tied up with all of our cultural understandings or is there something about intelligent behavior that might help us recognize it across scales and that's what turns us to these fundamentals like memory anticipation context dependent decision making and then once we open up to that functional level of intelligent behavior in cognition we start seeing it everywhere and so instead of just seeing these cells and their behavior as taking part of an intelligent system that again only blinks into existence at the top level they're in and of themselves intelligent actors awesome so let's see the video got me locked in here sorry how about that you want to take us through that slide daniel yep so it's written in the paper i propose oh well we'll come back to um homeostasis in a second at the top it's written ubiquitous exchange of slowly changing ion-based voltage signals within and among cells all cells are electrode electrically active and we talked about that just a few slides ago how ancient are these mechanisms how early was bioelectric coordination exploited by evolution so this is raising the notion that early cognition was biomechanical it was biophysical it involved morphological computation like we saw on the previous slide with the movie of cells but also electrical affordances existed from the beginning and so if biophysical affordances existed from the beginning and allowed for cognition to arise then it makes just as much sense to say that electrical affordances for cells to communicate and do these types of cognitive processes also were exploited very early in evolution so primitive meaning at a smaller level but also earlier on and suggesting that it's bioelectric was one of the earlier uh mechanisms of cognition is pretty cool and i like this quote from the paper that says the ability to operate toward a region and state space may be the primitive origin of complex cognitive systems that can entertain counterfactuals which is to remember or anticipate events that are not occurring right now so we'll touch back on homeostasis but how are the goals related to homeostasis how does bioelectricity play a role in homeostasis and i think we come back around to that on 17 i wanted to give an example about where bioelectricity and development came together with cognition and for some people genetic reductionism makes it so that you know the story is not complete until we know what

gene is involved and i think that they'd find this recent 2021 paper from drosophila fruit flies to be really interesting because what they do with a series of experiments is actually demonstrate the importance of membrane depolarization which is to say bioelectric capacity in uh gene regulatory networks so like hh and ptc are uh proteins that are produced by loci and dna and they're part of feedback loops involving the depolarization of membranes so they show that regulation of that membrane potential has an important role in signaling hedgehog and that there's a mutually reinforcing relationship between these two processes and then what does that allow well it allows morphological computation of this developing disk of the fly that later turns into the wing and so there's advantages and disadvantages like with diffusible molecules there's a tremendous amount of expressivity with what that molecule is like you have an infinite number of hormone possibilities but on the other hand it's very slowly diffusing and maybe you can only signal a short range on the other hand there's only like one electricity there's not like the blue membrane potential and you know the pink membrane potential as parallel membrane potential there's only one membrane potential so it's kind of a big knob to turn however as shown here it really does play a functional role in stabilizing gene regulatory networks if you were missing that part of the puzzle it would be as incomplete as if you were missing a critical signaling molecule so i suspect that as advances the kind that are going to be referenced in this paper uh advances in being able to visualize and track and manipulate bioelectric fields become more accessible that more and more systems that were thought to be purely chemical or purely mechanical i think that there'll be a more important role for bioelectricity right it is like one big surface one membrane potential so you have one membrane and that's like the the differentiator right between that's the boundary right is what is enclosed or encapsulated in the membrane versus what is not uh okay so here we are active inference this is the final keyword and i pulled this image out from the active inference lab um a previous uh live stream that we did with um ines and uh tomas or thomas so this is the free energy principle computation and computationalism and realism and tragedy and so this is just the basic action perception loop so you have active states external states sensory states and internal states and so the both of the active states and the sensory states are the blanket states right so you have the the internal can act on the external uh via these two blanket states so this is all of the sensory perception of the organism and the degree on which they're able to act on the environment do you want to add any thoughts here daniel it's just that active inference is a unifying framework for thinking about perception cognition and action and it does that in a scale-free way that's grounded in bayesian statistics and physics so it's going to be deployed as

a way to help us look for different kinds of behavior across scales because the internal state could be internal to a cell it could be internal to a tissue internal to the organism etc and it's that expressivity of the active inference framework that is going to come into play in the awesome and it's a process a scale-free process okay active inference as brought up in this paper the author says agents scaled up by evolving from basic homeostatic loops driven by active surprise minimization via addition of delays which are memory anticipation which is inference and networks spatially distributed processing that enables learning and progressive abstraction or generalization from the data gathering into larger collectives with optimal information structure also improves the computational ie predictive capabilities and gives rise to functional relationships such as memories encoded goal states test operate test exit loops that exist over and above any individual member these levels coexist enabling numerous coherent selves of different skills to be implemented by any collection of living matter that's intense uh the other one of the other pieces about active inference that differentiates it from a framework like reinforcement learning is right there um that there's the minimization of surprise and in order to minimize surprise through time you kind of have to be doing two things that at first pass don't seem like they're going to facilitate each other which is the pursuit of what you already know is going to be useful that's called pragmatic value as well as the pursuit of sampling things that will be informative that's epidemic value and so in active inference the way that agents through deep time minimize their surprise about perception that's what's referenced in the top paragraph about perceptual control control theory the way that the agents minimize their surprise about their future expectatio expectations of perception is going to be by jointly working on the pragmatic and epidemic aspects of their behavior and thinking about what behavior is going to be minimizing free energy in terms of kind of existing on the optimal trade-off where you're performing but also learning so these are a few features of active inference that differentiate it and then if we think about like a growth cone of a neuron is it doing reinforcement learning is it doing reward maximization it just barely makes sense but once we start to think about surprise minimization like it's endowed with an evolutionary prior it expects to connect to a neuromuscular junction then maybe some of its searching behavior makes more sense and in the top section also the author you know states that the sensory input layer is what enables active inference to operate so because there's receptors and channels that that's the perception the perception that for a cell um yeah anyway just thought i'd add that in okay greeting sarah hi sarah uh daniel do you want to read these uh core assumptions and i'll fix the thing thank you classic so i'm only gonna read the uh the two uh

highlighted parts here because these are paragraphs that people want more detail on they can definitely read the paper to learn more about there's three core assumptions that are laid out really clearly so i appreciated that about this that it makes clear assumptions and shows how those assumptions come into play and then basically there's two avenues for disagreeing i guess one would be i think those are invalid assumptions but they're um properly applied or one can think that there are valid assumptions but improperly applied and so that helps structure discussion with the contents of the so the first core assumption is a commitment to evolution every capacity has a natural history and emerged from simpler variants um i guess also evolution can perceive through a reduction of complexity so eventually it had to have emerged from simpler variants but in the short time i guess it could have emerged from more complex variants but the commitment to evolution means the commitment um to the dispelling of magical thinking and not looking for some sort of a metaphysical answer to our physical and the second assumption of the three is it is assumed that all metaphors are to be judged by their utility in driving scientific progress and there is not a binary categorization of scientific pictures which should be taken literally or not which can be decided a priori so we're not going around with um post-it notes saying what's real or what's not we're using utility as a guiding metric um we can actually just take a quick visualization and then sarah you can say hello and give any uh thoughts if you'd like but we previously in live stream 15 and talked a lot about this continuum from realism to instrumentalism like is active inference the territory itself that's realism we're really describing what's really there when we make an active inference model or instrumentalism on the right side here is like active inference as a map we're just using it as something that could be um used to capture some features of a system and then used instrumentally like a model and then here we've introduced sort of this third vector of utility which is like okay whether it's the map or the territory orthogonal to that is is it useful and so this is a space that's describing not just where we can think about active inference existing but also ways to evaluate different models not that one is better than the other but these are just different dimensions that models exist on all right okay maybe go to the third core assumption on to the third one so much of the discussion centers around goals related to teleology a hotly debated topic here goal directedness is taken in the non-magical cybernetic engineering and control theory sense of a feedback system that operates to maximize some specific state of affairs which can be modeled as a dynamical system with the tractors in its state space so to reiterate these three assumptions the first one is evolution the second one is utility and the third one is a non-magical sense of goal directedness so these

three assumptions play nice with each other and then they also lead directly to this final statement that except for a few brief remarks at the end no claims about consciousness defined rapidly as first person experience or sense of self as qualia are made all the examples concern functional third person objective capacities computation and behaviors so we're kind of like outside of systems looking at them as an investigator this isn't about the first person perspective and there's always so much fascination and interest in what consciousness is but in a way these three assumptions go a really long distance towards just saying hey let's look at the systems behavior as we observe it rather than as we imagined it to be like so that's at least the starting point for a great discussion um before we jump into the figures sarah do you want to give any thoughts or say hello uh sure yeah i am i i'm kind of coming in pretty disoriented but uh tried to follow michael's work uh quite a bit and i'm enjoying uh discussing it from whatever angle comes up and oh i don't know uh background who cares um i'm in philosophy right now um yeah that's fine nice well any slide you want to chime in on is great so find the slides okay they're yeah anyway um um go for figure one blue if you want sure so this is a continuum of cognitive powers so and this is just the first part of the figure we'll show the next part in the next slide so this is a tote which is test operate test exit loop uh which is a schema of a basic homeostatic cycle so the author says that by continuously taking action to minimize the distance the difference in error between the current state of affairs and a set point describing a different possible future state of affairs it enables a system to pursue goals despite perturbations from the outside world and intrinsic noise so this is i thought cool like you know the first like possible counterfactual like yes it's 75 degrees and i have plenty of sodium and potassium and you know calcium but but what if i don't always what if it's not always this way right what if what if my state changes so like that the first counterfactual maybe enabled this kind of homeostatic feedback loop i thought that that was a neat idea put forward in the paper so how is this related to active inference or systems engineering daniel well the terminology reminded me a lot more of engineering than we often see like sensors and actuators it makes it sound like it's going to be a robot and also this like test exit sounds like we're going to be doing sort of software evaluation but we can pretty clearly see the skeleton of action perception loops whether it's framed as it was a couple slides ago with internal external states and then blanket states or here where you have to basically have the difference between the desired and expected value leading to policy selection not shown here leading to action actions cause events in the environment on the yonder side of the blanket and then the state of the external environment influences the measurements on the inbound side which are again

contrasted with the idealized state and that leads to another round of action selection so this is a engineering way of laying out the test operate test exit loop but who knows we're able to jump from active inference pretty nicely into a more engineering frame and so perhaps engineering could be integrated closely into active inference yeah it seems like the blanket states like the sensors and actuators like our um you know the action perception part are here in the middle and so it kind of reminds me of the active inference loop like diagram that we showed earlier so it's interesting uh next part do you want to go through this one danielle yeah so in figure 1 b the caption reads that the scale is showing how different types of activity and systems can be ranked according to their degree of purpose-ness so again think about purpose in the third person not the first person it's not the sense of purpose but more what i believe hofstetter called the anti-species i'm not sure to pronounce it but it was like think about the most mindless behavior and then push away from that and that's what you can call purposeful rather than imagining that there's like this pure crystallized purposeful behavior and it sort of goes bottom to top in terms of the degree of purpose ability and on the bottom we see um basically just action that doesn't appear to have any kind of um sense or purpose to it like maybe gas molecules bumping around and then we progress through higher and higher levels of feedback loops and then each one of these brackets it's kind of like a distinction so like starting left behavior you can take the right branch or the bottom branch passive behavior like a ball that's being thrown or active okay active behavior is it gonna be non-random um or random like many things are random but you know is it a wind-up toy that's just doing something that is not not appearing to be too purposeful and you continue up and up the branch choosing the higher branch each time until you get to these first and second order predictions and higher order recursive logics which are seen as higher and higher levels of purpose cool do you have any thoughts sarah okay uh daniel do you want to take this what is an individual well there's so many ways to go about this and as a ant researcher that's always the debate like okay is the individual the six-legged next mate and then that makes the colony like a society or a super individual or is the colony the individual and then the nest mate is like the tissue so there's a lot of discussion around what constitutes an individual and how would you determine where the boundaries of individual individuality are and that piece i just referenced about establishing the bounds can be done in a few different ways one of them would be looking at where information theory is strongest or where information flows are occurring that's the information theory of individuality and here we're going to take a cognitive perspective on individuation and mccollum is quoted so early in the neural network

days the definition in the paper is i propose a definition of an individual based upon its information processing the scales and types of goals that a system can pursue defines slash determines the boundaries and contents of the putative agent so it's an informational lens on what constitutes an individual like we can say that individual people become maybe an individual mob just one mob not multiple mobs when they start integrating information differently or when they start pursuing different kinds of goals in a different way and it touches on these big questions like why does it matter how we define individual as well as what other ways of thinking about individuality exists yeah it's intense right so not only like in the information theory of individuality is it like the the most concentrated information but it's also like that bi-directional information flow right it was is how uh krakow and that group the defined individuality in that paper and so that's super interesting to me like at what point is there downward causation in an aggregate like so it's at at that point that is that is like the the crux for how they um talked about individuality which is i mean it makes sense to me right like you know my my brain tells my foot to kick the ball or i decide i want to kick the ball and so my body then acts accordingly but it's um it's really interesting to think about when that when that hap emerges that downward causation in a system anyway something i think a lot about probably too much what about you tara yeah that is actually one of the few papers i actually have read and i like went with a highlighter and everything um and i i always wonder on on i was looking through the math on that and i actually learned a lot about like that i didn't even think was possible yet but it was it was binary in a sense it was digital and and i'm never clear about when that matters and when it doesn't in these kinds of models so that's one thing that came up for me sounds good next no no it's not the sims it's um a cone of individuation and in 2a is um this sort of like cone-based representation of individuality that's going to be returning in a few different guises in and on the plane going out in different directions from a central point is like the zone of influence or the extent of the system's ability to perceive act plan and then piercing that coming up and down in the c dimension is going from the past to the future and this is a cool way to look at how multi-scale systems or what are called compound intelligences compound intelligences down there on the bottom right exist so we can use this kind of a cone representation or um to look at single cognitive agents and potentially contrast them like in part b where we have a tick with what's hypothesized to be like a smaller scope spatially in the now as well as a shorter temporal scope in the past and future compare tick with a human and then also ask maybe about what kinds of other cognitive forms exist and then these cones can be composed really nicely so like on the example on the

bottom right with an ant colony again controversial to say that the ant is the organism when potentially the colony is the organism that's why i usually call them an estimate but we can see how the nest mates are inside of the colony within this cone and so the colony pushes out the boundaries of what these ants can do so they're kind of like on different parts of the spatial plane in the current moment and then the colony is something that is cognitively able to move beyond the time horizon of an individual nest mate by virtue of its collective organization and information processing this is cool and something that we like talked about a lot like offline in preparation for this and preparing the slides is like the inversion of the um the typical like light cone so maybe daniel you want to walk us through this light cone and then we can maybe go back to the other one yeah so this is a similar light cone notion and it's just that instead of having um the two cups you know meeting mouth to mouth and then becoming pinched off in small time another way that light cones have been represented is actually with the current moment like the eye of the needle you know like a hourglass the pinch point and then in the past and in the future are the uh set of states that influence a given point and then the set of states it influences so let's like think about that butterfly flapping its wings in the here and now so if you go back one time step the air molecules that might influence it are like one layer of air molecule away and as you go back and back and back more and more spatial extents matters for the state of the butterfly in the and then similarly one time step in the future the butterflies flapping is only able to influence a small zone proximal to the insect whereas further and further out into the future more and more spatial extents might be modified because of the actions that happen in the here and now and then this is a leecors cabinet uh you know joke on the liquor cabinet but it's a paper that actually uses this kind of an approach to do modeling of empirical data and to fit machine learning models which we can talk about uh going forward like why and how do you get a speed up of certain kinds of simulations or data fitting by approaching it like a light cone so it's just interesting the axis here like and this is kind of how i'm used to seeing these represented are time and space right so like in physical space i am physically in this space that i am at right now and it's interesting as like you know as you go forward in time like you know given enough like hours i could theoretically like fly from new mexico to singapore but like for right now i mean i'm kind of limited to like my house or my block or you know like i don't have like point-to-point transport so i always like the the way that um mike represents this light cone like in this way like in the inverse it's it's like time and space and and you know it gets it starts in a very broad space and it proceeds to a very narrow space and it's like of course in the past like if i go to my distant past like i came

out of an egg that was in my mother's uterus right like so or in my mother's womb like that's where i came from i was in that one point and i've gone all kinds of places i guess until now but i'm not i don't feel like i'm many places now and and i also don't feel like i'm going to be many places in the future so i'm interested to to talk to mike when he comes onto the live stream about um the way that this is represented because it's very different you guys have any more thoughts on that yeah i have one um i mean it seems to me he's almost trying to do he's almost trying to take the case of the light cone and look at a particular ambition that you might have at this point in time and then and then can you carry out that ambition through time and as you get further and further you realize there's all these impediments you know that that come in your way and so it's this like one axis is the ability to envision um what it is you want to do and the other act yes this is the ability to enact it is the ability to well even control or whatever your environment any number of things that are required to enact uh this this vision that you have and that's an interesting interesting access transformation agreed um the light cone where the uh that looks more like an hourglass is kind of related to our limited degrees of freedom in the current moment but it you know expanding out into the future of what we can influence and then the the diamond shape that we see in figure two is more like our zone of what can be cognized which goes beyond what we can influence in the current moment because there's some future time when you won't be able to imagine you know you know as as the the curtains close on life and it narrows in and it's like smaller and smaller and then there's some point where your capacity to influence is actually kind of toned out and so there are two complementary ways to look at the same thing although it's interesting i want to say because you know like you look at like there could be some inner super meaning or intervening thing where it actually helps you that you never considered in this in this cone of possibility so you become wider like really it's just this the winding road right for sure all right moving on let's go to 28. this is sort of taking the light cone there's just one and a half more slides that take the light cone in a little bit of a different more physical direction but again it's it's all part of a continued conversation so we can look at the now that's the middle of those three layers where there's two different uh like individuals you know sally and uh you know friend n so it's like two of those are at different points now and so then we look over on the time cone on the right and so agent n is you know nick that's the one that we're focusing on and now the same time they're at the same moment but they're in different points in space and so from the point of view of n s is a space-like uh point but then in the same spot in the future or the past f or p in the same location you know the same address 20 minutes ago is type like in the same address

20 minutes in the future is also time-like but then on the edges of this cone is like light which i don't know the physics behind but i think it's kind of a cool idea that space and time have this boundary nature and that there's things that are more space-like like things that are instantaneously at the same moment but in different spaces and more time like the pure case being the same location at a different time but then it's like i'll meet you in 20 minutes two blocks away that's like moving through both space and time and maybe light does that in a kind of unique way so that was just one um way that in relativity that they talk about light cones and indeed the light cone concept is a physicist's concept that's really cool to think about like that light is the boundary between space and time that's kind of a neat thought yes and then on 29 i just wanted to pull one the details going beyond me and this discussion but here in a recent paper that explored how you could do simulations using light cones and so this is not just like sort of a metaphor or a cognitive heuristic this is something that you can use for large data analysis and so the way which i alluded to earlier in which you can speed up a simulation potentially by using a light cone is by having a point that you're focusing on and then you only need to consider one time point in the past at a radius of one spatially two time points in the past a radius of two spatially three to three out and so it helps you reduce the space of what you need to calculate from all versus all to the idea of calculating more and more out in more and more past steps and more and more out as you go into the future um so that's because in the physical interpretation there's a speed of propagation through a media so things don't happen instantaneously so that's like the light moving through a media at speed of light and then in the case of information propagation it's similar like if you wanted to focus on a given person and their role on an informational network you would maybe make like an informational cone instead of needing to calculate all by all at every single time steps and so it just made us curious about how active inference related to light cones so how can we apply this to some of the simulation models that we've already seen and then just you know what does the of the light cone represent what is the current moment what's the future just fun eternal questions cool ah that's so so i made this slide this is funny this light cone i actually showed um when i gave a seminar at mike levin's lab a couple years ago it's from um andrew gallimore's paper uh restructuring consciousness the psychedelic state in the light of integrated information theory but i was thinking about this in terms of shrinking and expanding the boundary of the self which the author discusses a lot in the paper here um and just like how that would change your effective light cone right like so if you have you have your your self boundary it's like a a limited like cone like the darker one that's shown in this figure on the

right and if you expand the boundary of yourself it expands your light cone so it made me think of that as we were talking as he was talking as mike was talking about the um shrinking and expanding the boundary of the self in the paper and so i'm really curious here's a couple quotes from the paper about um boundary establishment and maintenance so he says the surrounding body becomes an informational shield or markov blanket for the stem cell and this is he references carl firsten's 2013 paper um and so i wonder about like the markov blanket and i think that you know daniel this is something you and i have talked about as well about the markov blanket like being the defining boundary for the individual and like how i mean things must pop in and out of a markov blanket right like if i think about the cells that are in my skin layer right like i mean i'm losing them all the time and new ones are constantly being added so there's no distinct hard line to be drawn so like under like how do things pop in and out of a markov blanket like and what do you can you like just have like peripheral boundary nodes on the edges there like i've talked before also about like the difference between like your fingernail and your cuticle like there's like a a small tiny layer that doesn't know whether it's a fingernail or a cuticle so it's like how do you like have these like boundaries right um anyway another quote from the crucially by organizing into a partial electrochemical and informational pool or sync to tm sinks somebody say that for me please syncidium syncidium there we go all of the cells are able to measure and detect events occurring within the same boundary creating a larger individual that emerges from the collective and i think this quote was like in in reference to the gap junctions and how they expand that electrochemical pool by like they open up like some like they pull the plug on the electrochemical gradient and so it kind of enables this like cell linkage so that's one one way that um you know boundaries are maybe established but any thoughts on this guys we'll come back to it i think i have a few thoughts on where the markov blankets come into play through time cool all right uh so homeostasis which we we promised to come back to so what evolutionary pressures led cognitive boundaries to the to expand this is from the paper and the author proposes that the atom of this cognitive hierarchy is homeostasis um so reactive homeostasis evolves into predictive allostasis under the pressure to predict signals from the environment and other elements of the biosphere and that's like what i was talking about earlier about the first like counter factual about um you know what if it's not always 76 degrees and full of nutrients in my environment how can i regulate that reduce my uncertainty about future availability of the supplies that i need and also there's this um nice progression from homeostasis so that's sort of like similar you know same homeostasis it's the

thermometer is trying to get within the bound but the best way to stay within the bounds when the environment is changing is actually to not just be reactive but to be proactive or predictive and then what's the best way to be proactive well it's to have increasingly rich generative models of the niche if you know that night time is coming you're going to be able to take really good predictive behavior right now for tonight if you know that winter is coming at a larger time scale it's another level of you being able to make adaptive decisions so what looks like homeostasis sometimes can actually just be very uh skillful predictive allostasis [Music] do you have any thoughts sarah on that cool um okay so this one is memory and scale free cognition and active inference um so i don't know i just had some thoughts here like memory in as a gene regulatory network and this is kind of something that not really was stated in this paper but really like you know it's it's our evolutionary memory of like where we've been before and how we've um come to regulate and you know work within our environmental condition and like also memory is a process like i don't know i think about that like where is it stored is it externalized into the niche like we do a lot of niche modification and so i don't know um it's interesting to think about our our ram and our rom in um in like a biological sense so it's the author says a second step in the simplest homeostatic loop is the inclusion of a richer set of hidden layers in the neural network sense so these are additional biochemical nodes between sensors and effectors of a given system that enable a degree of memory and importantly memory can serve as the beginning of modularity because learning essentially groups diverse stimuli into compressed representations complex states of affairs become remembered as compact biophysical engrams this is the essence of the kind of modularity when a simple biophysical event kicks off the formation of a complex morphogenetic cascades such as building a hand in embryogenesis one nice point there is the difference between modularity and memory about how they enable each other it's sort of like if everything is being mixed in one cup there can't be a memory of different colors okay well if you have two separated cups you can remember two different colors and so on so memory can serve as the beginning of modularity is saying that it's this um relationship between things becoming distinct and being remembered through time distinctness carried through time is memory you can remember a longer number when you have more modularity that's being propagated through time and then the paper on the right is um a an older first in paper talking about memory attention and salience and active where they described uh memory working memory as a process of evidence accumulation in a temporarily structured hierarchy so i guess like going back through deep time like that's where the gene regulatory network kind of comes in

as a process right like the process of as and that's like the working memory there anyway and just you have one one note there on the multi-scale active inference models sometimes the way that memory is being modeled inactive inference is as a state variable that changes at a slower rate and so within the longer time scale model we have a shorter time scale model and then that can be modeled from setting to setting as memory carrying over and so it relates to this idea that the larger cognitive system is able to also act over longer time scales as well as reach back with a deeper memory because if you have the kind of short-term model that's just stand-alone it doesn't it's like a memory-less process but if that has inherited from a hierarchical model the state from a previous um click of the model it's memory and then if it's able to imagine one more click out into the future it's goal planning it's anticipation and so inactive inference one of the ways that we integrate memory and planning together is by thinking about hierarchically nested models so also i wonder what role memory plays in policy selection like i remember every summer when i go to scotland it is cold like i mean i live in new mexico it's like 120 degrees here fahrenheit so every time every summer i remember that when i go to scotland it's cold so now it's my policy to bring a jacket when i go right so even though it's the summer time and i wouldn't normally do that so is that memory also ever modeled in the policy selection daniel it's a good question like i wonder if you can plan to forget or you know i'll make a note i'll plan to remember i'd say active inference just gives us a way to think about all those different affordances that real cognitive systems have like offloading memory to the niche through external computation or in terms of what you just described but that's a good question we can ask right i don't actually even need to remember that it's cold in scotland because you know i can just ask google they will tell me the weather and it will be cold in scotland so if i just remember to you know check my cognitive offloading device then i don't even need to remember to break a jacket when i go so yep okay next so i figure i have one i have one random thought maybe it comes up in another slide but is this question that michael levin has about how does it know when to stop in terms of um cell growth and morphogenesis if it's the case that it's connected to memory in the way that he's suggesting then yeah you know like maybe if it related the information flow i guess that may be obvious given the focus of this um but it is interesting you know it's like the the cells are just like we've got we've got to now become multi-body you know or become social rather than individual in order to store memory like there's a funny like break point yeah i didn't see the stop mechanism really brought up in this paper but i have seen it so um i'm not really sure where it ties in um figure three i mean i mean if he calls it a morphogenic cascade you know

it's like well we're we we we're born we grow we grow to a certain point and then we shrivel and die so it's like how does that connect to memory and information ah it's kind of suggested yep so figure three is looking at a sort of cell's eye view of that time cone and now we're going to step away from you know is it a time diamond or is the time going and we're going to think about the trajectory of a given cell go back to the assumptions of the paper though this isn't about whether the cell is experiencing things it's not what it's like to be a cell it's about how the cell has a given spatial extent in a it has a given spatial extent that can be thought of as having a memory it inherits a history through time and then also has an anticipation zone in the future then it can do anticipation of memory as a sole agent or it can interact socially so what happens when it interacts socially with other agents like itself well it expands the spatial dimension in the now that's what's being called integrated spatial perception and that allows it to have expanded memory in the past like if each of your five friends have a book then there's more memory as a group because there's more books and then also the ability to extend the anticipation into the future and then this part i just highlighted in blue is joining into communication networks allows the cell to have temporally delayed access to the information obtained by neighboring this is where the time cone comes into play it's not that when something is perceived um by one of these cells off on the fringe the focal cell doesn't instantly hear about it they hear about it one time step later so that's where the times the time cone comes into play because from the point of view of this focal cell just assuming that um information moves one cell per unit of time then at zero time points its radius is one one time point away its ability to integrate sense is one more radius away and so you can keep on clicking back and that's reflecting a broader and broader zone of memory or you can carry it forward with a broader broader zone of influences or states that it can affect and so the reason the resulting larger individual and again individual is being used cognitively here not like evolutionarily is formed with a cognitive world that's unified by the cells sharing of information across time and space so no teleportation and no time travel the limitations of propagation of in our physical universe means that from the point of any given cell or agent it's existing in this time cone-like structure yeah that was like the expanding the boundary of the self um and like kind of like what happens like the time cone is expanded and you know what this just makes me think of well first i just want to remind everyone that this like cell cell communication is like what is what gap junctions facilitate because they're really connecting all cells and solid tissues um but but really i thought you know the old adage that you are your company and like if you think about you know your five like closest friends and that like

communication like network that you have like you hear about you know world events or events that are relevant to their lives it might be relevant to you like you hear about it one time step later and just how like you know your cognitive like boundary is expanded by like your inner circle i don't know made me think of that yep hey blue can you just can you just re-highlight or talk about the cell uh cell exemption that you're just mentioning oh yeah so that was earlier in the talk it was one of the key words so let me just go back to that slide um so here uh these gap junctions so they they are they were originally you know categorized in like neural tissue and uh in muscle you know to facilitate like rapid communication but they're really present in every cell in solid tissue and they they just perform um not ion exchange but it's like passive diffusion of ions so that it's like almost a shared intercellular space so would it be safe to call that a kind of a bottleneck i just wonder so maybe not as much of a bottleneck as it is like um it pokes a hole in the informational boundary yeah okay so that it's more the more rapidly let's just say that the gap junction so the no gap junction it moves like um takes three time steps for the signal you know one time step to release the molecule one for it to diffuse one for it to be perceived so every three units a time you're getting a radius expanding out by one now let's just say that a gap junction allows one time step the direct diffusion so then the time cone is expanding wider because the diffusion from the target cell is able to sweep out a larger time cone whereas if again if it took three units of time for every one of diffusion it'd be like a narrower time cup so rap communication which is never instant allows that time cone to access more into the present and then influence more broadly and so bottleneck is sort of like thinking like a bottleneck is a constraint relative to the width of the bottle but in the context of a gap junction it's actually like a channel or an opening for information transmission so what do you think i'm just i'm just gonna go ahead if you think about it in terms of like your um inner circle of friends imagine like you can wait for your friend to call you up and tell you this exciting new event or like you can have a wire tap into their house so like the gap junction is like having a wire tap into their house so like you always know like what is going on right okay i'm also searching for another analogy that's just saying into the stuff i've been looking at lately do you think it's fair that analysis to um you know a compression function like you know if you picture the classic neural net you know a bunch of nodes here just a few in the center of the hidden layer and then expanded out would would it better be a fair analogy for a gap junction yeah i'm not seeing it sorry okay okay i would just uh to recover it i would just say that the transmission is like a compressed form of it's like a little sampling of the chemical milieu from one cell being passed to the other if there was no

bottleneck no compression of information they would be just in one unit there'd be in one markov blanket and so there's this trade-off where if you totally isolate the rate of information propagation is low the modularity is very high though on the other hand with total integration you can't have memory because you don't have modularity however you do have instantaneous or close to that transmission and then the gap junction exists at this kind of sweet spot that allows for modularity with narrow channels of communication which is what shannon was studying in the first place cool so figure okay so figure three i think we just put popped on the cone is that true yeah that's just to draw the connection between the figure two with the the time diamond time cone and then just this figure cool okay figure four uh so figure three oh yeah well the the figure three the slide that we just had was like as things build out so that was the point of view of the agent building out and now it's like things fall apart so figure four we see a sort of rehearsal in b of what we saw previously which is that that target cell it's combining sensory information that's able enabling it to detect patterns more rapidly getting a big picture that's what it says in b and therefore missing information can be anticipated by this pattern seeking drive but what happens when things go wrong that's in c where this bottom right cell the green one has gone rogue it has stopped communicating in a way that was participating in this broader sense making and it has shrunk its sensory radius down and so in doing so it reduced the cognitive capacity of the other four cells it has reduced its own sphere of influence and it's also introduced this adversarial dynamic potentially over evolutionary or developmental time skills cool any more thoughts on that one reminds me of a friend i know but no other way it also reminds me about how cancer research is starting to think more about the niche and the signaling between cancer and non-cancer cells like cancer cells need to call down healthy blood vessels they need to prevent healthy immune response from taking them out and so it's like cancer as a dialectic or as a relationship between agents with different goals again not the experience of the goal but acting as if it has a goal that third person cybernetic teleology and so that helps us make sense of all these disparate features of cancer by thinking about internal and information propagation etc what do you think it would be right or off to think about the one possible place where that could go wrong um happening at the gap junction like the gap junction would no longer function as a good sampler compression thing which i know you don't agree with the analogy but something like that i like i i don't know what's happening in that research area but does that make any sense yeah we talked about gap junctions earlier and their involvement in um neoplasticity and all kinds of other disorders definitely i just found this really nice paper called gap junctions in cancer communicating for

50 years and it's the abstract starts with 50 years ago tumor cells were found to lack electrical coupling leading to the hypothesis that loss of direct intracellular communication is commonly associated with cancer progression and onset so that's pretty interesting it's like when you shut down the lines of communication like between two countries are those two countries having an improving relationship or is that like the last time that they spoke before war so closing down the embassy shutting down the gap junction it doesn't bode well for a sort of amicable resolution to conflict or communication that's why it's so important to maintain the avenues of communication even when there is an adversarial relationship because things can always get worse you can always shut down the communication which will make the generative models and the incentives and the goals diverge even more cool i don't know just all of a sudden you go into politics and i'm like oh yeah but wait a minute anyway no but but we should shut down they're bad we should show if they keep the channel open you also stand more chance of injury by way of keeping the channel open so keep the channel open until you shouldn't um this is just um a fun uh discussion for a little bit on the evolutionary anachronism which means like uh just something that's happening at a different time than it's expected like a feature from the past that gets brought back like an evolutionary reversion and so in the paper it's described that uh this is a breakdown of multicellularity and highlights the fact that the scale of the structure which cells work to maintain can change cooperation towards an entire organ system or body that's a healthy organism to the level of a single cell so instead of seeing cancer just mechanistically as a derangement of signaling or derangement of replication and dna reproduce reproduction of a cell we can think about it informationally and cognitively and cybernetically like the cell has changed its cybernetic horizon from the goals of the organism to its own local goals and so this concept of an evolutionary anachronism and how even in the current day we can study anachronism uh in examples with avocados a tasty fruit and as per the wikipedia it says avocados are exceptionally fatty fruits with seeds far too large to be successfully dispersed by any wild animal presently alive in the americas and so that is a paper of jansen and martin from the 80s which was actually talking about the ecology of large animal plant interactions of animals that we've never directly seen but it's like if you know that there is a really large seed and you know that there's seed dispersing mammals then it can lead you to hypothesize that there was a seed disperser that was large enough to eat avocados whole um and so we can look at anachronism in the current moments and help us you know retrace the past so just like the avocado is an anachronistic trait with a such large seed given to the current ecology past ecology we can see cancer as this reversion to a previous mode of

information integration and goal setting that it's like it's inappropriate in the context of the goals of the body not necessarily within the goals of a single cell that just wakes up one day it's like wow i'm bathed in glucose i should replicate so can't blame that stuff also like the cancer a lot of the markers in cancer cells are really like it's like a developmental reversion so not sure if you guys are aware of that or not but it's like they they revert to like a more immature cellular phenotype and so that i mean it's literally like going back to the past so it's interesting like is there something can be learned from the degree of reversion uh the degree of i'm not sure here's one thing that comes to mind there is there's some cancers where the the it goes back to like a stem cell of that tissue so like it's like a white blood cell cancer that like replicates a lot of white blood cells when there is a cancer in a reproductive organ sometimes it's like called a teratoma because it can have like hair and nails and stuff like that because it's actually pulled back so far that it can then differentiate on all the pathways rather than a reversion up from being like a healthy reproducing white blood cell precursor to just a higher like a white blood cell stem lineage but it doesn't pull back all the way to including ectoderm whereas if it was in um you know a reproductive cell then it could do that cool biofilms do you want to take this one danielle yep so i think the key point here is that bacteria have ion channels just like metazoans do you know dopamine gpcrs the proteins that signal between neurons those are not innovations of mammals they're not innovations of vertebrates they're those kinds of channels and diffusible molecules are cellular they're pan cellular and so what's being shown here is just like in previous figures we saw one kind of eukaryotic cell right you knew because it had a nucleus we had one eukaryote connecting with a few other eukaryotes well here we have one prokaryote connecting with other prokaryotes to do the exact same thing which is increase its sensory capacity again not instantaneously but think time cone like cone and so in the same way that the eukaryotic cells form a cognitive through their associations and affordances such as the gap junctions bacteria when they form a biofilm can also form a larger cognitive integrated and you can still say well but from an evolutionary point of view this the unit of selection is the single bacteria okay that's the evolutionary individuality definition and when we use the cognitive individuality definition we see a lot of the similarities between a multicellular organisms tissue and then a multicellular biofilm tissue even not in a biofilm i mean bacteria have an incredible ability to like i mean it's like antibiotic resistance right like they do plasmid exchange which isn't like their chromosomal dna but it's like small little snips of dna like they just like have a handshake like it's like a bacterial handshake like now i'm gonna give you some antibiotic resistance too like we spread germs

like covet and bacteria to one another they spread like plasmids to one another which which confer different properties you know and we select for this like antibiotic resistance through the use of all of our like antibacterials and so only bacteria that have these little plasmids are survive it's really interesting so even outside of a biofilm they do some some kind of um you know communication and like cognitive expansion of their self in like the way that we do with our inner circle of friends cool all right key ideas um i just highlighted a few of these like points but there's a huge box in the paper so we'll just read through the highlighted parts here i'll take this one so first a unified integrated cognitive self or individual can be defined with respect to the integrated ability to pursue specific goals via a homeostatic process that resists perturbations second an agent's cognitive world can be quantified and characterized enabling comparison with others regardless of their material implementation by estimating the spatio-temporal boundaries of its area of concern the volume in space and time over which the agent is able to take measurements exert influence and functionally link disparate events learning and association three the borders of the temporal and spatial events of which a given system is capable of measuring and acting map out a cognitive light cone a boundary in the informational space of a mind so estimating the spatio temporal boundaries of an agent is something i find particularly interesting so i just put that in as a question at the bottom do you want to read some of the next ones daniel yep so these are some key ideas four five six so four cancer is a reversible shrinking of the computational boundary of a biosystem pretty tantalizing there with the reversible i'm sure that's something that a lot of people would like to see maybe the idea is you remind the rogue agents about bigger goal like hey you know i know you're hungry but we wait in line at this restaurant so that everybody gets what takes care of them and maybe that helps remove the you know cancerous uh restaurateur five agents scaled up by evolving from basic homeostatic loops driven by active inference which is surprise minimization by the addition of delays which are memory anticipation which is inference inference has planning as inference and which is spatially distributed processing enabling learning and progressive abstraction generalization and data that's pretty cool to see active inference and elaborations of active immigrants at the heart of this future electrical evo eco diva it's like saying you start with action and perception and then you can carry that forward in time planning have an extended trail of memory that's you know the past and then you can also expand spatially but again a spatial isn't instantaneous there's a propagation speed in the media and so even laterally similar agents interact through this light cone way so nice to see where active inference plays into like cones there and then six infotaxis the drive for better

actionable intelligence about the regularities and patterns in the world and in the agent's own mechanism encourages cells to connect in groups via signaling this is another area where active inference has a lot to say and is a value add relative to a reward only framework like reinforcement learning because again what drives the bacteria to coalesce and to work together so maybe it's reward but then you have to appeal to this idea that they're maximizing reward with their behavior and the state or leave and all this kind of stuff info taxes helps us understand that the way to get good actionable intelligence which is kind of like active inference a i right there is to make your actions exist on the trade-off frontier like the pareto optimal trade-off between getting it done right now and expanding your epistemic boundaries sometimes the right move is to contact somebody who knows better or to go to a website that might inform you better and so it's actually a useful action in the moments to arrange to get better information later and so active inference helps us think about how these different needs are traded off by organisms cool all right so seven collecting into oh i can't do it syncidium sensitium do i do it okay collecting into a syncidium enables all of the cells to share the same data and access the same memories um that's pretty awesome you must assimilate it's like the borg yep sorry star wars nerd um okay also greed eight greed at the single cell level for information infotaxis drives cooperativity as each unit expands its measurement boundary communication with neighbors and thus inevitably becomes part of a bigger self with bigger set points serving as homeostatic attractors so i think active inference also has a role to play here in this cooperativity so like i can reduce my uncertainty about my by cooperating with people who have the same goal that i have right like or cooperating becoming part of a group you know and and therefore our efforts make a bigger effort or bigger anyway less surprise when i'm in a bigger group because it expands that that light cone right and then nine there is a fundamental symmetry between anatomical control mechanisms and cognitive mechanisms and um i think that this was discussed in terms of bioelectricity in the paper do you guys have anything to add one thought on that is cooperation arising at a higher level so that just reminds me of like if you're looking for a microscope and you saw a cell a eat cell b you'd go i mean adversarial relationship right could there be anything more directly adversarial and then you pull back and it's a white blood cell you know that's attacking a cancer cell so it's cooperative at a higher level but it can look adversarial at a low level and so what does that map onto in different systems how can we re-imagine conflict and tension as actually playing a stabilizing or cooperative role at a larger scale can interpersonal differences in perspective or disagreements lead to a cooperative or regenerative collective this is what we hopefully move

towards i saw some pigeons procreating yesterday while i was out walking around in my neighborhood and it definitely looked adversarial like it looked very very adversarial but they were cooperating at a higher level looks can be deceiving uh okay definitely i was thinking of a case where like if a tiger attacks me i should just assume that it's like an information transfer that needs to happen i don't know come get peace with it sarah oh it's like in um the selfish gene by dawkins he says uh yeah why would you go beyond the gene does it make any sense you know would you have mammals like uh doing kind acts for each other just look at the lion eating the gazelle and how could that possibly be a collaborative relationship and then i thought what if those two populations stabilize each other through deep time and they can't coexist and then they actually persist better both because of that one act so who's to say but just one thing like if that is all the case you know if this is all like whatever man the universe becoming like why would it be that that we had such a hard time with it like shouldn't it shouldn't there be some mechanism that allows me to just be like oh okay cool you know like i don't get that i think the mechanism is narrative cognitively it might play out differently in different systems but agree to disagree it's a narrative that we can both agree on we're cooperating at that narrative level doesn't mean that we don't disagree in fact we entrench that we disagree but we're agreeing at a higher level cool do you want to read the next ones so yeah just a few of the key points from 10 11 12. uh 10 bioelectric integration helped evolve control strategies and cognitive content across the continuum from chemical networks to human minds 11 the key points here were that there's a scale in variance with how different systems make decisions and that we can across different systems study how the cybernetic processes of learning and parameter optimization are implemented by a large number of units pursuing infotaxes and homeostatic goals aka pragmatic value and episodic value and 12. a conceptual unification is proposed in this paper as scale-free cognition and um then it's talked about how these boundaries between different subunits are a major control knob between the self and the world like the cell is doing the same stuff whether it's part of an organism or a part of a cancer but it's actually this control knob like what are the limits of the cognitive that sets the cell up to act a certain way or another and uh then a reminder at the very end from a utilitarian perspective that the most appropriate level of analysis is to be determined empirically aha was empiricism determined empirically and that is said the level that facilitates prediction and control with the least effort by the experimenter or by the system itself what if they disagree and gives rise to a unified understanding that drives the most novel robust research programs at the bench nice you know if it doesn't advance your career as a scientist that's not the case

this is you daniel active inference in multi-scale communication just wanted to capture a little bit about multi-scale systems and active so here we have nested systems so the top left you know that's an ant brain that's a picture that sasha took of an ant brain that came from a pogo ant and then there's the colony with a bunch of nest mates dragging food around and then the colony is embedded within an ecosystem so within one nested system you know within these time diamonds that are sort of nested inside of each other we can think about how the internal state of one is linked to the external state being the next level up like the internal state is the brain and the external niche to the brain is the body okay now the body and the brain are the internal state for what the colony and then the colony is like the internal state for the ecological niche so that's something that we've seen before with how active inference helps us understand and model and simulate hierarchically nested systems but also we can think about interactions at the same scale like collective behavior collective intelligence as occurring when at the same of agent they're interacting like laterally and so that's this teal arrow that's where you have two agents there are they're each other's external state so active inference helps us zoom up and out and down and in and then it also helps us go laterally and then not to repeat it too many times but the lateral transmission isn't instantaneous it's dynamic it happens through time which means that we don't need to look at all lateral interactions just the time cone cool uh predictions and research program uh there was a big list here so i just pulled a few of them out i'll read some of these off which are a lot of them questions so is there a quantifiable sense in which biological systems model themselves can an in silico evolutionary system be built containing both genetic and physiological components which simulates the scheme described above homeostasis and infotaxis and illustrates the emergence of different scales of cognitive horizons over time can we directly observe the evolution of multicellular goals from networking of agents with single cell homeostatic goals might there be higher level effects on the physiological boundaries of the cognitive sub-agents within the body from exposure to psychedelics which have long been claimed to expand transpersonal boundaries or induce ego death the analogies between geometric spaces and cognitive ones need to be explored further as has begun beginning to be done via geometric information theory time diamond anyone definitely i think that there's a lot a lot of room to go here will complex engineered systems set novel goals and if so how it is a prediction of this view that adaptive useful robotics will require components that are themselves competent and goal seeking that is super interesting also um because really as an intelligent you know being capable of cognition so are all of my parts so if we're going to really import you know give rise to intelligent life intel intelligent

artificial life we have to start with intelligent artificial components i think that's super cool and also goals we can't we can't just say well let's just build you know a bigger car or a bigger computer i mean what do we want this intelligence design system to be intelligent in the service of or in the way of and uh it's a disaster waiting to build capacity without goal for sure do you want to read some more of these yep so the second set of predictions so this is where mike is setting out a little bit of his uh agenda the next work in this subfield i just you know the subfield each of the fields they're modular they have their own disciplinary histories their own disciplinary anticipation but this subfield should focus on the discovery of bioelectric communication inducing technology so just like once we had chemical biology people were really excited about finding drugs that bound to proteins well now that we're aware of bioelectricity let's think about manipulations and measurements of bioelectricity two it's a prediction that it's possible to induce the formation of metazoan like bodies in an otherwise unicellular by forcing the expression of appropriate electrogenic proteins and gap junctions so he's trying to get towards some unique predictions and experiments uh prediction that transcriptomic which means gene expression analyses of regeneration events from bacterial biofilms plants and metazoan models of limb regeneration should show consistent use of electrogenic proteins in the events that enable cooperation towards self-limiting morphogenetic cascades so it's kind of the um the structure informational or cognitive structure of recovery and repair might have some similarities across systems that wouldn't immediately be seen as otherwise scale-free cognition hypothesis suggests that multihuman systems cough remote teams could have their own degree of cognition so as we always say you know things that the brain knows that the neuron doesn't things that the colony knows that the ants decimate doesn't there's got to be things that remote teams or any other kind of team know that individual participants don't know and then the exobiology question is given a novel life form or just a novel form how does one know when successful communication has taken place and that's an interesting question for a lot of you zeno slash exobiologist seti anyone cool next slide so this is in the section what it feels like to be a pancreas but i i really pulled um the point that kind of hit home for me the most out of the section and the final point that mike makes in so i'll just read this off it is striking that the process which zen practice is meant to reverse attachment to past memories and high valence for future expectations or fears is precisely the suggested to be responsible for the creation of complex selves with a capital s it is unclear whether it is beneficial or even possible to truly live in the moment and let go of past memories and future expectations but anyone who succeeded in doing this would achieve

precisely what zen promises the dissolution of the self and the final sentence of the paper which i just loved recreating the unification into and liberation from larger scale unifying unifying cells with a capital s would be a true pinnacle of synthetic biology and artificial intelligence engineering wow this is good paper um so just key questions at the end what might a good understanding of this work enable what are the unique predictions and implications of this work what are the next steps for fep and in relation to this work what are the goals of this research and what are you still curious about so we will be thinking of all of these things as we get ready to do um 25.1 and 25.2 over the next couple weeks so thank you for participating there is a feedback form for feedback and you can get in touch with active inference lab at all of these links on this slide nice work blue great job with your first broadcasting and facilitating on this one that's really fun so yeah anyone's welcome to participate or contribute to our live activities asynchronously synchronously we'll figure out the right performance for you so thanks sir also for joining thanks again blue